Omni-Prot: Unifying Protein Function Prediction with PLM-Centric Fusion

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Author's note: The authorial "we" is used as conventional scientific style; all research and writing are solely my independent work.

Abstract

Computational annotation of protein function remains severely constrained by the enormous gap between available sequences and experimentally validated characterizations. Current prediction methods often fail to fully exploit modern protein language models by compressing their representations too aggressively, depending on a single model, requiring auxiliary inputs like structural data and evolutionary alignments that limit applicability, or employing architectures that fail to encapsulate the full expressiveness of the features. We introduce Omni-Prot, a unified computational framework that can be tailored to diverse protein prediction challenges by strategically combining representations from four complementary protein language models: ESM-C, ProtT5, Ankh3, and PGLM. Our architecture employs reciprocal cross-attention mechanisms coupled with adaptive gating strategies to enable effective communication between model streams, producing integrated representations that capture both fine-grained sequence motifs and broader organizational patterns. Evaluated on major benchmarks including Gene Ontology annotation, enzyme classification, and stability regression, Omni-Prot establishes new performance standards, with especially pronounced gains in the most established functional categorization tasks. Systematic ablation experiments reveal that commonly relied-upon inputs such as three-dimensional coordinates and multiple sequence alignments contribute negligibly when powerful language model representations are properly leveraged, suggesting that large-scale self-supervised pretraining effectively encodes the critical biological signals these supplementary features provide. Furthermore, by depending exclusively on amino acid sequences, our approach demonstrates more robust generalization while simplifying deployment and eliminating error propagation from upstream computational pipelines. We provide open-source implementations and an accessible web interface to facilitate widespread adoption in fields including but not limited to therapeutic development, protein design, and functional genomics research.

1 Introduction

Proteins are the molecular workhorses of life, orchestrating virtually every biological process from catalyzing reactions to transmitting signals and maintaining cellular structure. Understanding protein function and properties from amino acid sequence alone remains one of the most fundamental challenges in biology, with direct implications for drug discovery, synthetic biology, and disease research. While experimental characterization provides ground truth, it is dwarfed by the sheer scale of available sequence data: high-throughput sequencing has cataloged ~246 million protein sequences in UniProt as of 2024[1], with metagenomic studies discovering 1.17 billion novel protein sequences that show no similarity to reference genomes, grouped into over 106,000 new protein families[2]. Yet the vast majority of these sequences remain without precise functional annotation, and experimental characterization is limited to ~574,000 manually curated entries in Swiss-Prot (less than 0.25% of the total)[1]. This massive gap between sequence availability and functional understanding, spanning Gene Ontology annotations, enzyme classifications, and biophysical properties, creates a critical bottleneck in translating genomic data into actionable biological insights[3,4].

Consequently, developing computational methods for automated protein function prediction has become essential to bridge this widening gap, enabling biologists to rapidly identify proteins of interest and guide virtual screening and engineering efforts[3–5]. Traditional approaches have relied primarily on homology-based inference, transferring annotations from experimentally characterized proteins to similar sequences using tools such as BLAST and profile-HMM pipelines (e.g., Pfam/HMMER)[6–8]. More recently, machine and deep learning methods have emerged that learn latent relationships between sequences and functions, often surpassing simple homology transfer on modern evaluations[9,10].

However, three key limitations persist across these approaches that our work addresses. First, prevailing systems underuse protein language models (PLMs): they compress rich embeddings through aggressive low-dimensional bottlenecks, rely on a single PLM despite complementary inductive biases across models, or employ architectures that fail to fully express the rich features the PLMs provide, thereby leaving much performance untapped[11]. Second, the field frequently misattributes what actually drives gains, overemphasizing structural inputs above all, and placing undue weight on other factors such as multiple sequence alignments (MSAs) or other auxiliary features[10,12,13]. Third, many architectures introduce input modality dependencies that complicate deployment: requiring structural coordinates, MSAs, or specific features that add computational overhead, increase inference latency, and risk propagating errors from upstream pipelines—all while amino acid sequences remain the universally available and most reliable input modality across the entire protein universe.

In this work, we present Omni-Prot, a unified framework that addresses these limitations through a single, PLM-centric architecture capable of tackling a broad spectrum of protein learning challenges with minimal task-specific redesign. Our approach fuses representations from multiple supporting protein language models. It employs bidirectional cross-attention combined with gated pooling mechanisms and clever training strategies to create a rich, multi-scale representation that captures both local motifs and global sequence properties. This fusion strategy exploits the complementary strengths arising from each PLM's distinct training objective, architectural design, and pretraining data: ESM-C[14] leverages efficiency-scaled masked language modeling on diverse protein sequences to yield high-quality, compute-efficient representations;

ProtT5[15] transfers T5-style span-denoising objectives from NLP, trained on UniRef sequences, to capture both local motifs and long-range dependencies through its encoder-decoder architecture; Ankh3[16] enriches purely sequence-based features via multi-task pretraining that combines MLM with sequence completion objectives across broad protein families; and PGLM[17] employs a unified autoencoding and autoregressive framework, bridging understanding and generative priors. By strategically combining these models with their varying inductive biases, the fused representation covers broader and more nuanced biological signals than any single PLM alone.

Across all major benchmarks tested, Omni-Prot sets new performance standards, delivering particularly important improvements over prior work for Gene Ontology term prediction (molecular function, biological process, cellular component) while outperforming task-specific models and complex multi-modal architectures. Our molecular function ablation studies reveal surprising findings about commonly used components, showing that 3D coordinates and MSAs provide negligible benefits when strong PLM representations are available; demonstrating that large-scale self-supervised learning in PLMs can single-handedly capture the essential signals most supplemental features provide that are needed for accurate and generalizable protein learning. In addition, our ablation experiments further reveal that integrating multiple PLMs, rather than relying on a single model, consistently yields superior performance, highlighting that complementary language models capture distinct aspects of sequence biology whose combination drives the strongest predictive gains. By incorporating bidirectional cross-attention pooling and bidirectional transformer blocks[18], our architecture enables the PLMs to communicate effectively and integrate their complementary strengths for each task.

Beyond accuracy improvements, we prioritize practical deployment and accessibility. Crucially, by relying solely on primary sequence, the framework avoids dependencies on unreliable 3D coordinates, low-quality MSAs, or other auxiliary features, ensuring more robust generalizability, even for poorly characterized proteins where such inputs are unavailable. We release our implementation as open-source software along with a web application that provides instant predictions across all supported tasks, enabling researchers and practitioners in drug development, protein engineering, and pathology to leverage state-of-the-art function prediction capabilities that substantially outperform currently popular web tools like BLAST. The result is a single, adaptable protein representation learning framework that unites rigor with broad utility for a comprehensive array of objectives.

2 Tasks, Datasets, and Evaluation

We evaluate our proposed framework across five objectives encapsulating three established protein learning tasks: Gene Ontology term prediction, Enzyme Commission number classification, and stability regression. This diverse benchmark suite encompasses multiple learning paradigms—including both classification and regression problems, hierarchical and flat label structures, varying degrees of label imbalance and long-tailed distributions, and data-scarce versus data-rich scenarios—providing a comprehensive evaluation testbed.

2.1 Gene Ontology Term Prediction

Task background. The Gene Ontology (GO) provides a controlled vocabulary for describing protein function across three orthogonal ontologies[3]: molecular function (MF), biological process (BP), and cellular component (CC). MF captures the specific biochemical activities a protein can perform (e.g., ATP binding, oxidoreductase activity); BP describes the broader pathways and objectives in which a protein participates (e.g., DNA repair, signal transduction); and CC localizes the protein to subcellular structures or complexes (e.g., cytosol, ribosome). Accurate computational GO prediction from sequence enables systematic functional annotation at scale, directly supporting drug discovery through identification of therapeutic targets and off-target effects, advancing precision medicine by interpreting the functional consequences of genetic variants found in patient genomes, accelerating enzyme engineering by identifying promising candidates for industrial biocatalysis, and facilitating comparative genomics by revealing functional conservation and innovation across the tree of life. Beyond annotation, GO predictions guide experimental prioritization by generating testable hypotheses about uncharacterized proteins and enable pathway-level analysis that connects individual protein functions to systems-level cellular behaviors. Not to mention, the task is technically challenging: labels are hierarchical, exhibit long-tailed frequency distributions, and require models to generalize to remote homologs and rare terms while remaining ontology-consistent. Note that this task was the central task targeted in this work and is regarded as one of the most important benchmarks in protein informatics since it is well-defined while addressing the key overall aspects of protein function through its three ontologies[19].

Dataset. We use PDBch[20], a curated dataset from prior work of 36,641 protein chains from the Protein Data Bank (PDB) clustered at 95% sequence identity, retaining one representative chain per cluster only if it has at least one GO annotation in MF, BP, or CC. The dataset is split 8:1:1 into 29,313 training, 3,664 validation, and 3,664 test proteins, with 489 MF terms, 1,943 BP terms, and 320 CC terms represented in the label space. Training and validation sets exclude proteins without labels in the target ontology, while the test set remains fixed across all three ontologies for standard cross-study comparability. This construction ensures that all evaluation proteins have ground-truth annotations and that the model is assessed on its ability to predict functional terms within established ontological categories. The dataset exhibits moderate label imbalance typical of real-world annotation databases, with rare terms having 50 positive examples and common terms appearing in thousands of proteins.

Evaluation metrics. We employ three standard metrics to assess prediction quality[21]. GO predictions undergo standard hierarchy-aware propagation preprocessing to ensure consistency with the GO ontology tree:

1. **Protein-centric $F_{\max}$.** Measures the best thresholded trade-off between per-protein precision and recall; higher means balanced, reliable hits at the protein level.

2. **Function-centric $\mathrm{AUPR}_{\mathrm{macro}}$.** Averages average-precision over GO terms with positives so rare and common labels count equally; higher means better performance on long-tailed terms.

3. **Semantic distance $S_{\min}$.** Finds the threshold minimizing information-weighted error on the GO DAG, penalizing missed specific truths and spurious specific predictions; lower values indicate more semantically accurate and hierarchy-consistent predictions.

$F_{\max}$	$\mathrm{AUPR}_{\mathrm{macro}}$	$S_{\min}$
$F_{\max} = \max_{\tau \in \mathcal{T}} \frac{2\,\overline{P}(\tau)\,\overline{R}(\tau)}{\overline{P}(\tau) + \overline{R}(\tau)}$	$\mathrm{AUPR}_{\mathrm{macro}} = \frac{1}{\lvert \mathcal{L}^{+} \rvert} \sum_{\ell \in \mathcal{L}^{+}} \mathrm{AP}_{\ell}$	$S_{\min} = \min_{\tau} \sqrt{\mathrm{RU}(\tau)^2 + \mathrm{MI}(\tau)^2}$

2.2 Enzyme Commission Number Prediction

Task background. The Enzyme Commission (EC) number is a hierarchical, four-level taxonomy that specifies the chemical transformation catalyzed by an enzyme, organized by reaction class, subclass, sub-subclass, and substrate specificity[4]. For context, an identifier such as `EC 3.2.1.4` reads left to right as hydrolase → acting on glycosyl compounds → glycosidases → a specific substrate–reaction pairing. Predicting EC numbers directly from amino-acid sequence accelerates discovery in enzymology, synthetic biology, and metabolism. While still technically a multi-label classification problem like GO term prediction, the label cardinality is dramatically lower, and most proteins carry exactly one EC code, so in practice EC behaves like a near single-label task with an extremely long-tailed frequency distribution over specific codes.

Dataset. We follow the established EC-number benchmarking protocol from prior work. Specifically, the training data is derived from a UniProt-based enzyme corpus containing over 220,000 sequences. From this, a set of 74,487 sequences is obtained by clustering at 70% sequence identity, which we adopt directly for model training[22]. For validation, we use NEW-392[23], a collection of 392 Swiss-Prot enzymes released after April 2022 and spanning 177 distinct EC numbers, to guide model selection and early stopping. Final performance is evaluated on Price-149[24], an experimentally validated test set of 149 enzymes. See Limitations for information on extreme label imbalance and noise present in the dataset.

Evaluation metric. For EC prediction we report AUC (area under the receiver operating characteristic curve), measuring class separability where higher values denote stronger discrimination.

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d(\mathrm{FPR}), \quad \mathrm{TPR} = \frac{TP}{TP + FN}, \quad \mathrm{FPR} = \frac{FP}{FP + TN}.$$

2.3 Stability Regression

Task background. Protein stability is the ability of a sequence to retain its native fold under stress (e.g. heat, denaturants, proteolysis), typically summarized by a continuous score. Modern benchmarks frame stability as learning a mutational landscape around a few "parent" proteins: large libraries of single- and few-mutation variants are assayed to yield per-variant stability measurements. Predicting these scores directly from sequence enables data-driven protein engineering—identifying stabilizing substitutions, designing thermostable enzymes, and improving developability. Unlike GO or EC (classification), stability is a regression task with a scalar target. The core challenge is epistasis and landscape ruggedness: combinations of mutations can have non-additive effects, so models must generalize beyond single-mutation trends.

Dataset. We follow the TAPE stability task[5], which ships with 53,416 training, 2,512 validation, and 12,851 test sequences. The dataset originates from a high-throughput *de novo* miniprotein design campaign[25], where sequences were generated over four iterative design rounds and experimentally assayed for folding stability via yeast-display protease susceptibility. The benchmark trains and validates on design-round variants and tests on Hamming-distance-1 neighborhoods around 17 top candidates, probing local mutational effects on stability.

Evaluation metric. We report Spearman's rank correlation coefficient ρ, which quantifies the monotonic agreement between predicted and true stability scores; higher values indicate stronger correlation.

$$\rho = \frac{\mathrm{Cov}(r(\mathbf{s}),\, r(\mathbf{y}))}{\sigma_{r(\mathbf{s})}\sigma_{r(\mathbf{y})}} = 1 - \frac{6\sum_{i=1}^{n}(r(s_i) - r(y_i))^2}{n(n^2-1)}.$$

3 Model

Omni-Prot follows a general PLM-fusion architecture that can be easily adapted to a wide range of protein function prediction tasks, with independent model weights for each task. It begins by extracting per-residue embeddings from four complementary protein language models, which are processed through a dual-stream cross-attention encoder and bidirectional transformer. The contextualized representations are then compressed using sequence-level pooling with masking to handle variable lengths, and the resulting vectors are merged through a gated fusion mechanism to produce a unified protein-level representation. The bidirectional cross-attention mechanism enables each PLM to selectively query and integrate complementary features from one another, with learned gates dynamically balancing their contributions for the prediction task. A prediction multi-layer perceptron (MLP) head processes the final output features from the model to cover the relevant training objective (see Section 5.3 for architectural validation).

3.1 Inputs and Preprocessing

The input to our model is the raw amino acid sequence of a protein. Given a raw amino acid sequence $\mathbf{s} = (s_1, \ldots, s_L)$, each of the four complementary protein-language models m independently tokenize the input sequence with its respective tokenizer and generate per-residue features after the forward pass. Any auxiliary tokens introduced by the PLMs are replaced with model-exclusive beginning-of-sequence (BOS) and end-of-sequence (EOS) tokens, which act as registers for downstream fusion in our model[26]. Importantly, when sequences exceed the PLM's token limit, they are partitioned into overlapping windows and reassembled into a full-length representation through our sliding-window embedding strategy, ensuring complete coverage and utility of features regardless of sequence length.

$$\mathbf{H}^{(m)} = \mathcal{E}_m(\mathbf{s}) \in \mathbb{R}^{L\times d_m}, \qquad m \in \{\text{ESM-C}, \text{ProtT5}, \text{Ankh3-XL}, \text{PGLM}\}.$$

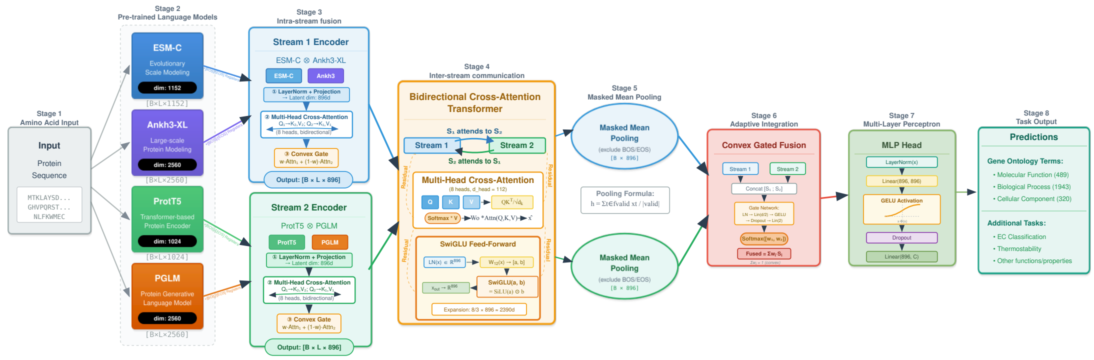


Figure 1: Omni-Prot dual-stream architecture. Intra-stream encoders first fuse PLM pairs (ESM-C, Ankh3-XL; ProtT5, PGLM) to form per-stream group representations; these encapsulated representations then interact in a bidirectional cross-attention transformer, followed by masked mean pooling, adaptive gated fusion, and an MLP head to produce GO/EC/stability outputs. Source: Graphic created by the student researcher using Inkscape, 2025. **Due to LaTeX SVG limitations, zooming into this diagram is highly recommended for visual clarity.**

Because the hidden dimensionalities of the four PLMs differ, all embeddings are projected into an optimal, validated latent dimension of 896. This not only aligns dimensions but also improves regularization by condensing representations to a common basis without aggressive condensation. For each $\mathbf{H}^{(m)}$, we apply normalization, dropout, and linear projection (no bias) in sequence:

$$\mathbf{Z}^{(m)} = \mathrm{Dropout}\big(\mathrm{Proj}(\mathrm{LN}(\mathbf{H}^{(m)}))\big) \in \mathbb{R}^{L\times d}.$$

Here, LN denotes layer normalization, Dropout applies stochastic regularization, and Proj is a learned linear map into the shared latent dimension d. The resulting projected embeddings $\mathbf{Z}^{(m)}$ serve as the aligned token-level representations for subsequent cross-attention pooling with BOS/EOS auxiliary tokens included.

3.2 Dual-Stream Encoder

We construct two token-aligned streams from the four PLM features: (ESM-C [600M], Ankh3-XL) and (ProtT5, PGLM [3B-MLM]). These paired streams are fused via pre-norm, convex-gated cross-attention, condensing the initial projected grouped latent embeddings $(\mathbf{Z}^{(i)}, \mathbf{Z}^{(j)})$ into a joint representation that blends attended features per token (amino acid) from both PLM feature sets:

$$(\,\widetilde{\mathbf{Z}}^{(i)},\,\widetilde{\mathbf{Z}}^{(j)}\,) = \Big(\mathrm{MHA}\big(\mathrm{LN}(\mathbf{Z}^{(i)}),\,\mathrm{LN}(\mathbf{Z}^{(j)}),\,\mathrm{LN}(\mathbf{Z}^{(j)})\big),\ \mathrm{MHA}\big(\mathrm{LN}(\mathbf{Z}^{(j)}),\,\mathrm{LN}(\mathbf{Z}^{(i)}),\,\mathrm{LN}(\mathbf{Z}^{(i)})\big)\Big),$$

$$\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\Big(\frac{\mathbf{QK}^{\top}}{\sqrt{d_h}}\Big)\mathbf{V},\ \ \mathrm{MHA}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{Concat}_{r=1}^{H}\Big[\mathrm{Attn}\,\big(\mathbf{QW}_r^{Q},\,\mathbf{KW}_r^{K},\,\mathbf{VW}_r^{V}\big)\Big]\,\mathbf{W}^{O}.$$

A small gating MLP produces tokenwise weights that sum to one, yielding a task-adaptive convex mixture:

$$[w_t^{(i)},\,w_t^{(j)}] = \mathrm{softmax}\big(\mathrm{MLP}(\widetilde{\mathbf{Z}}_t^{(i)} \oplus \widetilde{\mathbf{Z}}_t^{(j)})\big), \qquad \mathbf{F}_t = w_t^{(i)}\,\widetilde{\mathbf{Z}}_t^{(i)} + w_t^{(j)}\,\widetilde{\mathbf{Z}}_t^{(j)}.$$

Applying this module to (ESM-C, Ankh3-XL) gives the sequence stream $\mathbf{X} \in \mathbb{R}^{L\times d}$, and applying this to (ProtT5, PGLM) gives the sequence stream $\mathbf{Y} \in \mathbb{R}^{L\times d}$. Standard key-padding masks are utilized in attention operations.

3.3 Bidirectional Cross-Attention Transformer

After we have the features from each of the PLM groups, we engineer a bidirectional cross-attention transformer[18] over N stacked layers to exchange learned information between the two streams. Each layer $\ell \in \{1, \dots, N\}$ performs symmetric bidirectional attention: the (ESM-C, Ankh3-XL) stream $\mathbf{X}$ attends to the (ProtT5+PGLM) stream $\mathbf{Y}$, and vice versa, both using pre-norm multi-head attention followed by residual connections:

$$\widetilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell-1)} + \mathrm{MHA}\left(\mathrm{LN}(\mathbf{X}^{(\ell-1)}), \mathrm{LN}(\mathbf{Y}^{(\ell-1)}), \mathrm{LN}(\mathbf{Y}^{(\ell-1)})\right),$$
$$\widetilde{\mathbf{Y}}^{(\ell)} = \mathbf{Y}^{(\ell-1)} + \mathrm{MHA}\left(\mathrm{LN}(\mathbf{Y}^{(\ell-1)}), \mathrm{LN}(\mathbf{X}^{(\ell-1)}), \mathrm{LN}(\mathbf{X}^{(\ell-1)})\right),$$

where $\mathbf{X}^{(0)} = \mathbf{X}$ and $\mathbf{Y}^{(0)} = \mathbf{Y}$ denote the initial fused streams from the dual-stream encoder. Each attended representation is then refined through a residual feed-forward network with SwiGLU activation[27]:

$$\mathbf{X}^{(\ell)} = \widetilde{\mathbf{X}}^{(\ell)} + \mathrm{FFN}_{\mathrm{SwiGLU}}\left(\mathrm{LN}(\widetilde{\mathbf{X}}^{(\ell)})\right), \qquad \mathbf{Y}^{(\ell)} = \widetilde{\mathbf{Y}}^{(\ell)} + \mathrm{FFN}_{\mathrm{SwiGLU}}\left(\mathrm{LN}(\widetilde{\mathbf{Y}}^{(\ell)})\right).$$

The SwiGLU activation function provides enhanced expressivity through gated linear units with smooth, non-monotonic activation. Given input $\mathbf{h} \in \mathbb{R}^d$, the feed-forward network computes:

$$\mathrm{FFN}_{\mathrm{SwiGLU}}(\mathbf{h}) = \left(\mathrm{SiLU}(\mathbf{h}\mathbf{W}_1) \odot \mathbf{h}\mathbf{W}_2\right)\mathbf{W}_3,$$

where $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{d\times d_{\mathrm{hidden}}}$ project to an intermediate dimension $d_{\mathrm{hidden}} \approx \frac{8d}{3}$ (chosen to match parameter count with standard GELU expansions), $\mathbf{W}_3 \in \mathbb{R}^{d_{\mathrm{hidden}}\times d}$ projects back, $\mathrm{SiLU}(x) = x \cdot \sigma(x)$ is the sigmoid linear unit (Swish[28]), and $\odot$ denotes element-wise multiplication. This gating mechanism allows the network to selectively modulate information flow based on input content, improving parameter efficiency compared to typical GELU-based feed-forward layers.

Standard key-padding masks are utilized in attention operations throughout all layers. After N layers, the contextualized token-level representations $\mathbf{X}^{(N)} \in \mathbb{R}^{L\times d}$ and $\mathbf{Y}^{(N)} \in \mathbb{R}^{L\times d}$ encode complementary cross-modal dependencies in relation to the targeted task, ready for sequence-level pooling.

3.4 Protein-Level Fusion and Prediction MLP

After the final layer, we aggregate each stream with masked mean pooling that excludes padding and BOS/EOS registers:

$$\mathrm{Pool}(\mathbf{X}) = \frac{\sum_t \overline{M}_t\, \mathbf{X}_t}{\sum_t \overline{M}_t + \epsilon} \in \mathbb{R}^d, \qquad \mathbf{x}_{\mathrm{pool}} = \mathrm{Pool}(\mathbf{X}^{(N)}),\;\; \mathbf{y}_{\mathrm{pool}} = \mathrm{Pool}(\mathbf{Y}^{(N)}).$$

A sequence-level convex gate mixes stream embeddings similar to the function used in the dual-stream encoder:

$$[\alpha, \beta] = \mathrm{softmax}\big(\mathrm{MLP}(\mathbf{x}_{\mathrm{pool}} \oplus \mathbf{y}_{\mathrm{pool}})\big), \qquad \mathbf{z} = \alpha\,\mathbf{x}_{\mathrm{pool}} + \beta\,\mathbf{y}_{\mathrm{pool}} \in \mathbb{R}^d,$$

and a task-specific MLP produces the final output for the respective task:

$$\mathrm{MLP}(\mathbf{z}) = \mathbf{W}_2\,\mathrm{Dropout}\big(\mathrm{GELU}(\mathbf{W}_1\,\mathrm{LN}(\mathbf{z}))\big) \in \mathbb{R}^C.$$

The MLP consists of a layer normalization followed by a linear projection $\mathbf{W}_1$, a GELU activation[29], dropout regularization, and a final projection $\mathbf{W}_2$. This MLP is structurally identical to the one used in the convex gating formulation.

3.5 Training Techniques

Loss functions. For multi-label classification tasks (GO term prediction across MF, BP, and CC ontologies, as well as EC number assignment), we employ Asymmetric Loss[30] (ASL), a focal-style objective designed to address extreme class imbalance and long-tail distributions common in protein function annotation. ASL applies asymmetric down-weighting to easy examples while preserving gradient flow for hard positives, introducing probability clipping to prevent model collapse on trivial samples. ASL was found to improve convergence and performance compared to traditional Binary Cross-Entropy (BCE) loss.

Given input logits $\mathbf{x} \in \mathbb{R}^C$ and binary targets $\mathbf{y} \in \{0, 1\}^C$, the loss is computed as:

$$\mathcal{L}_{\mathrm{ASL}}(\mathbf{x}, \mathbf{y}) = -\sum_{c=1}^{C} \left[y_c \cdot \log(p_c^+) + (1 - y_c) \cdot \log(p_c^-) \right] \cdot w_c,$$

where $p_c^+ = \sigma(x_c)$ is the predicted probability for class c, $p_c^- = (1 - \sigma(x_c) + \delta)$ is the clipped negative probability with $\delta = 0.05$, and $\sigma(\cdot)$ denotes the sigmoid function. The asymmetric focusing weight is:

$$w_c = (1 - p_c^+ \cdot y_c - p_c^- \cdot (1 - y_c))^{\gamma_c}, \qquad \gamma_c = \gamma_{\mathrm{pos}} \cdot y_c + \gamma_{\mathrm{neg}} \cdot (1 - y_c),$$

with $\gamma_{\mathrm{pos}} = 0$ to preserve strong gradients for rare positive labels and $\gamma_{\mathrm{neg}} = 4$ to suppress easy negatives.

For stability regression, we use mean squared error:

$$\mathcal{L}_{\mathrm{MSE}}(\mathbf{x}, \mathbf{y}) = \frac{1}{N} \sum_{i=1}^{N} (x_i - y_i)^2,$$

where x_i and y_i are the predicted and ground-truth stability values, respectively.

Optimization. All models are trained using AdamW with decoupled weight decay. We apply a unified learning rate of $\eta = 10^{-4}$ across all parameters, with weight decay $\lambda = 0.01$, momentum coefficients $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and numerical stability constant $\epsilon = 10^{-8}$. Gradient norms are clipped at 1.0 to prevent instability during training.

Learning rate schedule. We employ a cosine annealing schedule with linear warmup[31] to stabilize early optimization and ensure smooth convergence. Let T denote the total number of training steps and $T_{\text{warmup}} = 0.05T$ the warmup period. The learning rate at step t is:

$$\eta_t = \begin{cases} \eta \cdot \frac{t}{T_{\text{warmup}}}, & \text{if } t < T_{\text{warmup}} \quad \text{(linear warmup)}, \\ \eta \cdot \frac{1}{2}\left(1 + \cos\left(\pi \cdot \frac{t - T_{\text{warmup}}}{T - T_{\text{warmup}}}\right)\right), & \text{otherwise} \quad \text{(cosine decay)}. \end{cases}$$

This schedule linearly increases the learning rate from zero to η over the first 5% of training, then decays it following a half-cosine curve to zero by the final step.

Exponential moving average. To improve model robustness and generalization, we maintain an exponential moving average (EMA) of model weights with decay rate $\tau = 0.999$. The EMA parameters θ_{EMA} are updated at each training step t via:

$$\theta_{\text{EMA}}^{(t)} = \tau \cdot \theta_{\text{EMA}}^{(t-1)} + (1 - \tau) \cdot \theta^{(t)},$$

where $\theta^{(t)}$ are the current model parameters. EMA accumulation begins after the warmup period (approximately the first two epochs) to avoid averaging poorly initialized weights. At inference time, the EMA parameters replace the standard parameters for all predictions. EMA parameters were found to greatly improve model smoothness at inference compared to non-EMA parameters.

Early stopping. To further prevent overfitting, we monitor validation performance with early stopping. For multi-label tasks (GO and EC prediction), we track function-centric $\text{AUPR}_{\text{micro}}$ with patience of 2 epochs and minimum improvement threshold $\Delta_{\min} = 0.003$. For stability regression, we monitor validation Spearman's rank correlation with identical patience and threshold criteria. Training terminates if the monitored metric fails to improve beyond $\Delta_{\min}$ after the patience threshold.

4 Results

We compare the performance of our model, Omni-Prot, against state-of-the-art baselines and benchmark leaders in their respective categories. We provide context for each compared model to highlight any specialized features or curated inputs employed. When considering the baselines at a glance, it is important to reiterate that Omni-Prot achieves superior performance while using only amino acid sequences and being a general architecture, not relying on auxiliary features such as MSAs, structural coordinates, and evolutionary profiles or task-specific objectives/architectures. This demonstrates a key finding supported by our ablation studies in the next section: PLMs can effectively capture the information provided by auxiliary features, and learning to extract and fuse representations from PLMs is of utmost importance in protein representation learning generally.

4.1 Gene Ontology Results

We evaluate on the PDBch test set and report $\text{AUPR}_{\text{macro}}$ (higher is better), F_{max} (higher is better), and semantic distance S_{min} (lower is better) for MF/BP/CC. Out of the three tasks considered in this paper, PDBch is the most important and well-studied for practical applications. Our model attains significant state-of-the-art results across MF and CC on all metrics and improves BP ranking quality ($\text{AUPR}_{\text{macro}}$) and semantic error ($S_{\text{min}}$), while remaining competitive on BP F_{max}.

	Molecular Function (MF)			Biological Process (BP)			Cellular Component (CC)		
Method	$\text{AUPR}_{\text{macro}}$	F_{max}	S_{min}	$\text{AUPR}_{\text{macro}}$	F_{max}	S_{min}	$\text{AUPR}_{\text{macro}}$	F_{max}	S_{min}
BLAST	0.136	0.328	0.632	0.067	0.336	0.651	0.097	0.448	0.628
FunFams	0.367	0.572	0.531	0.260	0.500	0.579	0.288	0.672	0.503
DeepGO	0.391	0.577	0.472	0.182	0.493	0.577	0.263	0.549	0.550
DeepFRI	0.495	0.625	0.437	0.261	0.540	0.543	0.274	0.613	0.527
PFresGO	0.602	0.692	0.417	0.293	0.568	0.535	0.361	0.674	0.498
HEAL-PDB	0.571	0.691	0.401	0.259	0.565	0.540	0.342	0.655	0.501
GGN-GO-PDB	0.601	0.698	0.400	0.291	**0.643**	0.546	0.347	0.657	0.510
Omni-Prot (ours)	**0.658**	**0.743**	**0.342**	**0.342**	0.615	**0.498**	**0.453**	**0.728**	**0.420**

Table 1: GO results on PDBch test set. Best values in **bold**. BLAST and FunFams are classic transfer baselines (not trained on PDBch). Among learned methods, DeepGO/DeepFRI/PFresGO use PDBch+SMch, while HEAL and GGN-GO are exclusively PDBch. Source: Table created by the student researcher, 2025. Performance metrics taken from Mi et al.[32]

Summary. On MF, Omni-Prot improves $\text{AUPR}_{\text{macro}}$ from the best prior (0.602) to **0.658** ($\Delta = +0.056$; +9.35%), raises F_{max} from 0.698 to **0.743** ($\Delta = +0.045$; +6.50%), and reduces S_{min} from 0.400 to **0.342** ($\Delta = -0.058$; −14.4%). These gains in molecular function prediction directly translate to more accurate identification of specific enzymatic activities and binding capabilities, beneficial for drug target identification. On BP, we lift $\text{AUPR}_{\text{macro}}$ from 0.293 to **0.342** ($\Delta = +0.049$; +16.7%) and lower S_{min} from 0.535 to **0.498** ($\Delta = -0.037$; −7.0%); F_{max} remains competitive (0.615 vs 0.643; $\Delta = -0.028$). The substantial $\text{AUPR}_{\text{macro}}$ improvement indicates markedly better ranking of rare biological processes, suggesting the model captures complex pathway-level relationships more effectively despite BP's inherent difficulty in threshold-dependent metrics. On CC, Omni-Prot advances $\text{AUPR}_{\text{macro}}$ from 0.361 to **0.453** ($\Delta = +0.092$; +25.4%), F_{max} from 0.674 to **0.728** ($\Delta = +0.054$; +8.05%), and S_{min} from 0.498 down to **0.420** ($\Delta = -0.078$; −15.7%). The dramatic $\text{AUPR}_{\text{macro}}$ gain reflects substantially improved discrimination of rare subcellular compartments and protein complexes, particularly valuable for understanding cellular organization in understudied organisms. Across ontologies, the consistent $\text{AUPR}_{\text{macro}}$ improvements demonstrate superior handling of long-tailed label distributions, while semantic distance reductions indicate predictions that are biologically coherent even when not exact matches, reducing the risk of misleading functional assignments.

Baselines at a glance.

- **BLAST[6]:** Transfers functions by nearest-neighbor local sequence alignment; reported as a classical homology-transfer reference.
- **FunFams[33]:** Groups sequences into CATH functional families to transfer annotations at the domain level using functionally coherent homology-based families.
- **DeepGO[34]:** An early deep model that learns sequence and PPI features while explicitly modeling Gene Ontology hierarchical dependencies.
- **DeepFRI[20]:** Utilizes a graph neural network over residue–residue contact maps combined with sequence embeddings from a protein language model to predict GO terms.
- **PFresGO[35]:** An attention-based method that integrates GO graph inter-relationships with representations from protein language models.
- **HEAL[21]:** Applies a hierarchical graph transformer with contrastive learning on structure graphs with node PLM features for protein function prediction.
- **GGN-GO[32]:** A geometric graph network with node embeddings from PLMs that captures multi-scale structural features (atomic and residue levels) with geometric vector perceptrons—the previous state-of-the-art, relying heavily on structural inputs at the expense of fully leveraging sequence-based foundation models.

4.2 EC Number Prediction

We report AUC (higher is better) for Price-149. We note extreme label imbalance and some label inconsistencies between training and testing splits that may affect absolute performance metrics (discussed further in Limitations). Nonetheless, relative comparisons across methods remain informative, and Omni-Prot achieves the strongest results.

Method	AUC
GrAPFI	0.5095
ECPICK	0.5888
CLEAN	0.7334
GraphEC	0.8404
Omni-Prot (ours)	**0.855**

Table 2: EC results on Price-149. Best in **bold**. All models were trained on identical data.
Source: Table created by the student researcher, 2025. Performance metrics taken from Song et al.[22]

Summary. Against the strongest prior, GraphEC (0.8404), Omni-Prot attains an AUC score of **0.855** (Δ = +0.0146; +1.74%). Gains are larger versus earlier baselines: CLEAN (0.7334) $\rightarrow$ Δ = +0.1216; +16.58%, ECPICK (0.5888) → Δ = +0.2662; +45.21%, and GrAPFI (0.5095) → Δ = +0.3455; +67.81%. While modest in relative terms, this gain at high AUC values indicates fewer ranking errors among difficult enzyme pairs, especially notable due to the lack of curated, dependent features compared to other priors.

Baselines at a glance.

- **GrAPFI[36]:** Constructs an enzyme domain-similarity graph from InterPro features and propagates labels through graph learning to predict EC numbers.
- **ECPICK[37]:** Applies evidential deep learning to produce EC predictions with uncertainty estimates, improving trustworthiness on large databases.
- **CLEAN[23]**: Uses contrastive learning on PLM representations with positive and hard-negative pairs to annotate enzymes and identify mislabeled or promiscuous enzymes.
- **GraphEC[22]**: A geometric graph model that combines ESMFold-predicted structures with PLM embeddings, localizes active sites, and employs label diffusion to refine EC predictions.

4.3 Stability Regression

We evaluate on the TAPE Stability test set, reporting Spearman's ρ (higher is better). Omni-Prot achieves the strongest correlation by a significant margin.

Baselines at a glance.

- **k-NN[38]:** Non-parametric nearest-neighbor regression on sequence-similarity features (Levenshtein distance for proteins).
- **One-Hot[5]:** MLP head over one-hot residue encodings.
- **LSTM[5]:** Pretrained recurrent sequence model fine-tuned for stability regression.
- **RT[38] ($\mathcal{L}_{SC}$):** Regression Transformer with a self-consistency objective.
- **Transformer[5]:** Pretrained self-attention encoder fine-tuned for stability regression.
- **UniRep[39]:** mLSTM pretrained on large unlabeled sequence corpora, then fine-tuned for stability prediction.
- **ProteinBERT[40]:** Protein-specific BERT-style pretraining with supervised heads.

Method	**Spearman ρ**
One-Hot	0.19
k-NN	0.21
LSTM	0.69
RT ($\mathcal{L}_{SC}$)	0.71
Transformer	0.73
UniRep	0.73
ProteinBERT	0.76
Omni-Prot (ours)	**0.85**

Table 3: Stability results on TAPE test set. Best in **bold**. All models were trained on the same data splits. Source: Table created by the student researcher, 2025. Performance metrics taken from Born et al.[38]

Summary. Relative to the strongest prior, ProteinBERT (0.76), Omni-Prot reaches a ρ of **0.85** (Δ = +0.09; +11.8%). Gains vs other widely used baselines are consistent: Transformer (0.73) → Δ = +0.12; +16.4%, LSTM (0.69) → Δ = +0.16; +23.2%, RT$_{\mathcal{L}_{SC}}$ (0.71) → Δ = +0.14; +19.7%, k-NN (0.21) → Δ = +0.64; +305%, and One-Hot (0.19) → Δ = +0.66; +347%. These improvements indicate more faithful ranking across the mutational landscape, especially beyond single-mutation trends, supporting base potential for protein engineering efforts to identify stabilizing modifications.

5 Ablation Studies and Analysis

To validate our architectural decisions and understand the individual contributions of key components, we perform comprehensive ablation studies. MF serves as our main ablation benchmark due to its relatively balanced label distribution, moderate dataset size, and stability compared to other sparser tasks. We use $\text{AUPR}_{\text{micro}}$ as our primary metric for ablations because it provides a reliable, aggregate measure of overall system performance by pooling predictions across all classes, making it well-suited for isolating the impact of architectural changes without the variance introduced by rare term performance (though other metrics like $\text{AUPR}_{\text{macro}}$, F_{max}, and S_{min} generally fell in line with $\text{AUPR}_{\text{micro}}$). All ablations use identical training procedures unless otherwise specified, with compared models trained for the same number of epochs and monitored on the validation set of PDBch (defined in Section 2.1). The results in this section overall illustrate the sheer predictive power of PLM embeddings and their potential when coupled with faithful feature extraction mechanisms.

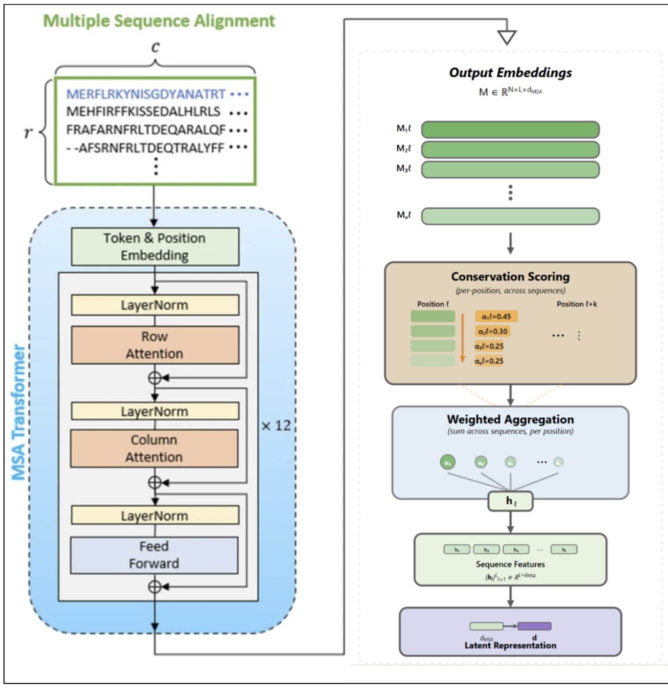
5.1 Impact of Auxiliary Input Features

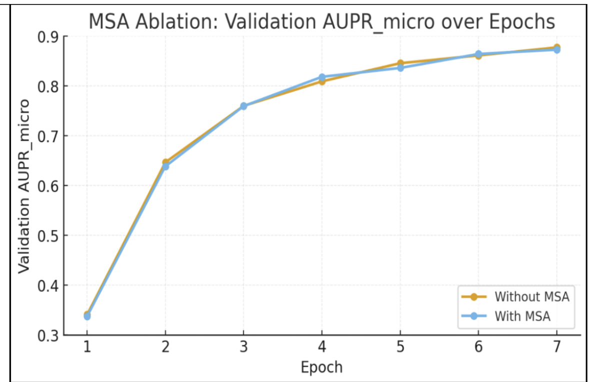
A prevailing assumption in protein function prediction is that auxiliary features like multiple sequence alignments (MSAs) and especially 3D coordinates (C_α positions) are needed to capture evolutionary conservation, structural context, and geometric constraints that raw amino-acid-based PLMs cannot alone provide. However, MSAs and 3D coordinates have significant limitations: MSAs require expensive homology searches, 3D coordinates are often unavailable or less accurate when relying on costly structure prediction, and both increase model complexity and can propagate upstream errors. Moreover, these inputs are unreliable for *de novo* proteins that lack evolutionary history and distant homologs that produce sparse alignments and less confident structure predictions. While other modalities utilized in literature could in principle be tested, we focus on MSAs and 3D coordinates as the two most widely adopted and information-rich sources of evolutionary and structural context. We therefore evaluate whether these commonly used inputs provide measurable gains over strong protein language model representations alone.

5.1.1 Multiple Sequence Alignment Ablation

MSA generation. For proteins available in the OpenFold dataset[41], we retrieved pre-computed MSAs by concatenating hits from three sequence databases (BFD+UniClust30, MGnify, UniRef90) and applied greedy Hamming diversity-maximizing subsampling to retain phylogenetically diverse sequences while controlling MSA depth. For proteins absent from OpenFold, we generated MSAs using ColabFold's MMseqs2-based search protocol[42] against the ColabFold database with modified parameters (90% sequence identity threshold, 75% coverage requirement, maximum 256 sequences). This two-tier strategy ensured comprehensive MSA coverage across our benchmark while maintaining consistent quality standards.

MSA encoding. MSAs were encoded using MSA Transformer[43], a 12-layer transformer pretrained on clustered UniRef50 alignments via masked language modeling. The model employs axial attention, factorizing full pairwise attention over rows (sequences) and columns (positions) independently, enabling efficient





Epoch	1	2	3	4	5	6	7
w/o MSA	0.341	0.646	0.760	0.809	0.845	0.861	0.877
with MSA	0.337	0.638	0.759	0.818	0.836	0.864	0.872
Δ	−0.004	−0.008	−0.001	+0.009	−0.009	+0.003	−0.005

Figure 5.1 | MSA ablation overview. Left: illustration of the MSA encoder and pooling mechanism. Right: validation AUPR$_{\text{micro}}$ across epochs for models trained with and without MSA features. Incorporating explicit evolutionary context yields only marginal gains, indicating that pretrained sequence models already capture much of the information encoded in MSAs.

Source: Leftmost MSA Transformer block adapted from Hong et al. (2022), "S-Pred," *Scientific Reports*. https://doi.org/10.1038/s41598-022-18205-9. All other panels created by the student researcher in Draw.io, 2025.

processing of large alignments while capturing both positional conservation and sequence coevolution. To integrate MSA representations into our architecture, we randomly subsample each alignment to 16 sequences and apply attention-weighted pooling that learns sequence importance scores based on evolutionary conservation. Let $\mathbf{M} \in \mathbb{R}^{N \times L \times d_{\text{MSA}}}$ denote the MSA embedding for N sequences of length L. The pooling mechanism computes conservation-based attention weights:

$$s_{n\ell} = \mathbf{w}^\top \mathbf{M}_{n\ell}, \qquad \alpha_{n\ell} = \frac{\exp(s_{n\ell}/(\tau\sqrt{N}))}{\sum_{n'} \exp(s_{n'\ell}/(\tau\sqrt{N}))}, \qquad \mathbf{h}_\ell = \sum_{n=1}^{N} \alpha_{n\ell} \mathbf{M}_{n\ell},$$

where $\mathbf{w}$ is a learned conservation scoring vector, τ is a learned temperature parameter, and $\mathbf{h}_\ell \in \mathbb{R}^{d_{\text{MSA}}}$ is the pooled representation at position ℓ. The resulting sequence-level MSA features $\{\mathbf{h}_\ell\}_{\ell=1}^{L} \in \mathbb{R}^{L \times d_{\text{MSA}}}$ are projected to the shared latent dimension d and utilized as an input stream.

Experimental setup. We trained two architecturally identical models on MF with different input configurations: (1) the baseline model using four PLM embeddings (ESM-C, ProtT5, Ankh3-XL, PGLM), and (2) an MSA-augmented variant where the pooled MSA features replace PGLM as the fourth input stream while retaining ESM-C, ProtT5, and Ankh3-XL. This substitution design ensures that both models have identical capacity and architecture, with four input streams fused through the same dual-stream cross-attention mechanism, isolating the effect of evolutionary information versus single-sequence language modeling. Both models were trained with identical hyperparameters and evaluated on MF over the first seven epochs to assess early learning dynamics and convergence behavior.

Results. The model without MSA features achieves comparable performance to the MSA-augmented variant throughout training, with final validation $\mathrm{AUPR}_{\mathrm{micro}}$ of 0.877 versus 0.872 respectively. This result falls in line with recent work[13], challenging the assumption that evolutionary information from MSAs is necessary when modern protein language models are available for pertinent protein function prediction. We believe this result is in part due to the fact that modern PLMs themselves are pretrained on millions/billions of protein sequences—internalizing co-evolutionary patterns.

5.1.2 Three-Dimensional Coordinate Ablation

Graph construction and architecture. We construct residue contact graphs using k-nearest neighbors on C_α coordinates with $k = 16$, recomputed per protein to capture local geometric context. Each node is initialized with ESM-C per-residue embeddings $\mathbf{h}_i^{(0)} \in \mathbb{R}^{1152}$, which are projected to a hidden dimension of 512. We employ an E(n)-equivariant graph neural network[44] (EGNN): node coordinates transform equivariantly under rotations, translations, and reflections, while the pooled graph representation is invariant to global pose. Unlike message-passing networks that operate only on node features, EGNNs jointly update node embeddings and coordinates through geometry-aware message passing.

Each EGNN layer updates node features and coordinates via:

$$\mathbf{m}_{ij} = \phi_e\Big(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \|\mathbf{c}_i^{(\ell)} - \mathbf{c}_j^{(\ell)}\|_2^2\Big) \qquad \mathbf{h}_i^{(\ell+1)} = \mathbf{h}_i^{(\ell)} + \phi_h\Big(\mathbf{h}_i^{(\ell)}, \sum_{j\in\mathcal{N}_k(i)} \mathbf{m}_{ij}\Big)$$

$$\mathbf{c}_i^{(\ell+1)} = \mathbf{c}_i^{(\ell)} + \sum_{j\in\mathcal{N}_k(i)} \frac{\mathbf{c}_i^{(\ell)} - \mathbf{c}_j^{(\ell)}}{\|\mathbf{c}_i^{(\ell)} - \mathbf{c}_j^{(\ell)}\|_2 + \epsilon}\, \phi_c(\mathbf{m}_{ij})\,.$$

Here, $\mathcal{N}_k(i)$ denotes the k-nearest neighbors of node i in coordinate space, $\mathbf{c}_i^{(\ell)} \in \mathbb{R}^3$ is the coordinate of residue i at layer ℓ, and ϕ_e, ϕ_h, ϕ_c are learned MLPs. The edge messages $\mathbf{m}_{ij}$ depend on squared Euclidean distances $\|\mathbf{c}_i - \mathbf{c}_j\|_2^2$, making the network sensitive to geometric relationships while preserving E(n) equivariance. Coordinate updates use normalized relative directions (with a small ϵ for stability) and learned scalar weights $\phi_c(\mathbf{m}_{ij})$, allowing the network to refine spatial positions based on feature similarity. After $L = 4$ layers, we apply a layer normalization to the node features and then aggregate via masked mean pooling to produce a graph-level representation:

$$\mathbf{g} = \frac{1}{N_{\mathrm{valid}}} \sum_{i=1}^{N} \mathbb{I}[\mathrm{mask}_i]\, \mathrm{LN}\Big(\mathbf{h}_i^{(L)}\Big), \qquad N_{\mathrm{valid}} = \sum_{i=1}^{N} \mathbb{I}[\mathrm{mask}_i]\,.$$

This protein-level embedding is then passed through an MLP classification head to produce final predictions.

Experimental setup. We trained EGNN models on all three GO ontologies (MF, BP, CC) under two conditions: (1) Real coordinates, using experimentally determined C_α positions from PDB structures[45]; and (2) Random coordinates, where each protein's coordinates were replaced by an i.i.d. random point cloud of the same length (no structural signal), after which the k-NN graph was constructed exactly as in the real

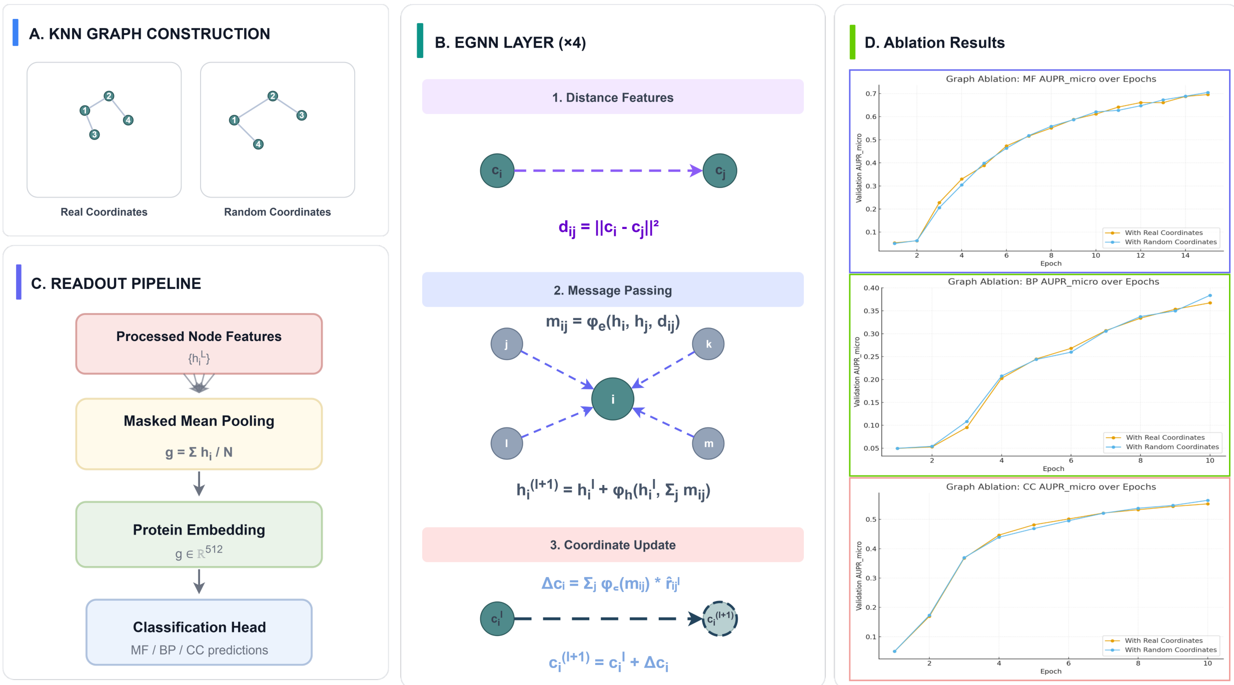


Figure 2: Coordinate ablation overview. (A) kNN graphs from real vs. random coordinates; (B) EGNN layer (distance, messages, coordinate update) ×4; (C) readout: masked mean pooling → protein embedding → classifier; (D) validation $\text{AUPR}_{\text{micro}}$ over epochs for MF/BP/CC comparing *With Real Coordinates* vs. *With Random Coordinates*.
Source: Graphic created by the student researcher in Draw.io, 2025.

case. Both conditions use the same sequences, masks, architecture, hyperparameters, and training budget, so model capacity and parameter count are effectively identical. This comparison isolates the contribution of geometric information in the coordinates themselves.

Results. Across all three ontologies and throughout epochs, models using random coordinates achieve nearly identical performance to those using real structures (e.g., MF $\text{AUPR}_{\text{micro}}$: 0.696 real vs. 0.705 random; BP $\text{AUPR}_{\text{micro}}$: 0.367 real vs. 0.374 random; CC $\text{AUPR}_{\text{micro}}$: 0.553 real vs. 0.565 random). This striking result demonstrates that geometry-aware message passing, even with a sophisticated equivariant architecture designed explicitly to leverage 3D structure, provides no measurable advantage when strong sequence representations are available. The similar performance between real and random coordinates suggests that the EGNN architecture was mostly reliant on the overwhelmingly feature-rich node embeddings from ESM-C (vastly outshining the 3D geometry). We hypothesize that ESM-C and related PLMs, trained on massive sets of protein sequences with diverse structural contexts, have internalized high-level spatial proximity patterns necessary for multi-label protein function prediction through co-occurrence statistics, effectively learning which residues appear together in functional motifs regardless of their linear sequence separation. This finding is further supported by evidence that large-scale continual structural pretraining of ProtT5 (as in ProstT5[46]) yields marginal improvements or even decreases in performance over the original sequence-only model (ProtT5) in similar function-based benchmarks.

5.2 PLM Scaling Ablation

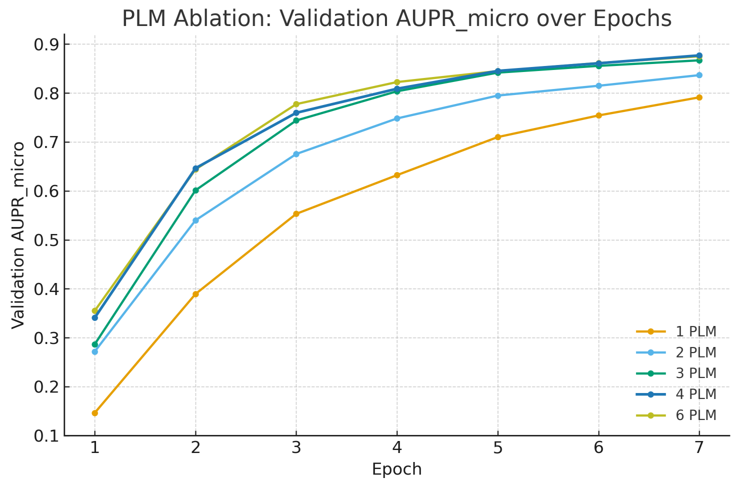


Figure 3: PLM Ablation: Validation $\text{AUPR}_{\text{micro}}$ over Epochs and # of PLMs. Source: Graph created by the student researcher, 2025.

Epoch	1	2	3	4	6
1	0.146	0.272	0.287	0.341	0.355
2	0.389	0.540	0.601	0.646	0.644
3	0.553	0.675	0.744	0.760	0.777
4	0.632	0.745	0.804	0.809	0.822
5	0.710	0.795	0.842	0.845	0.845
6	0.754	0.815	0.856	0.861	0.861
7	**0.791**	**0.837**	**0.867**	**0.877**	**0.875**

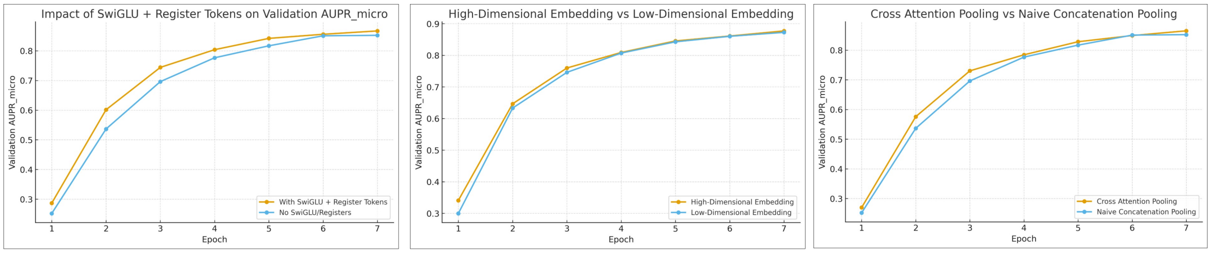
Table 4: Validation $\text{AUPR}_{\text{micro}}$ by epoch vs. # of PLMs. Source: Table created by the student researcher, 2025.

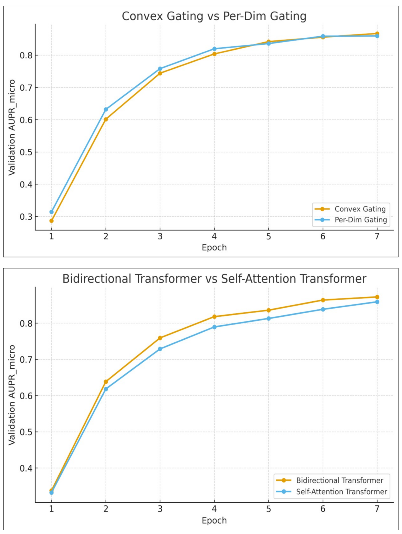
When using protein language models, nearly all prediction methods rely on a single PLM to extract sequence features. To determine whether incorporating multiple PLMs improves learning, we systematically vary the number of input representations and measure their impact on early-stage convergence.

Experimental setup. We test whether adding additional PLM streams improves early learning on MF through the first seven epochs. Starting from an ESM-C baseline, we incrementally add PLMs in the order: ProtT5, Ankh3-XL, PGLM, and the pair (MSA Transformer+GENIE[47]). The fusion architecture depends only on the number of PLMs: with a single PLM it reduces to a self-attention encoder; with two PLMs it becomes a bidirectional cross-attention encoder; with 3+ PLMs, additional streams are grouped on the two sides of the dual-stream encoder so that two aggregated representations feed the same bidirectional transformer. All variants hold the latent dimension and fusion modules fixed across conditions to control capacity.

Results. Adding PLM streams consistently improves early learning performance and convergence speed up to a threshold. As we increment from one to four PLMs, $\text{AUPR}_{\text{micro}}$ increases steadily across the first seven epochs, with each additional stream accelerating convergence to higher performance plateaus. This trend aligns with the complementarity hypothesis: each PLM contributes distinct inductive biases shaped by its architecture, pretraining corpus, and training objective. However, beyond four PLMs (i.e., when adding the MSA Transformer+GENIE pair for a total of six streams), performance gains stagnate and convergence rates plateau. This saturation likely reflects two factors. First, the pool of high-quality protein language models is limited, and models added beyond the top four may offer weaker representations or be optimized for tasks (such as generative modeling in GENIE) that do not directly support discriminative feature extraction. Second, representational redundancy likely increases with the number of PLMs: beyond four streams, additional models may largely recapitulate information already captured by earlier ones, yielding diminishing returns. These results suggest an optimal "sweet spot" of 3-4 complementary PLMs for multi-modal protein representation learning in the current landscape of PLMs.

5.3 Architectural Design Choices





Epoch	1	2	3	4	5	6	7
Bidirectional Transformer vs Self-Attention Transformer							
Bidirectional Transformer	0.3372	0.6382	0.7592	0.8180	0.8358	0.8638	**0.8724**
Self-Attention Transformer	0.3318	0.6180	0.7288	0.7895	0.8129	0.8383	**0.8589**
Δ (A−B)	+0.0054	+0.0202	+0.0304	+0.0285	+0.0229	+0.0255	**+0.0135**
Cross Attention Pooling vs Naive Concatenation Pooling							
Cross Attention Pooling	0.2695	0.5756	0.7300	0.7841	0.8282	0.8490	**0.8649**
Naive Concatenation Pooling	0.2518	0.5364	0.6960	0.7764	0.8167	0.8505	**0.8521**
Δ (A−B)	+0.0177	+0.0392	+0.0340	+0.0078	+0.0114	-0.0015	**+0.0128**
SwiGLU + Register Tokens							
With SwiGLU + Registers	0.2866	0.6012	0.7439	0.8036	0.8418	0.8556	**0.8669**
No SwiGLU/Registers	0.2518	0.5364	0.6960	0.7764	0.8167	0.8505	**0.8521**
Δ (A−B)	+0.0348	+0.0649	+0.0479	+0.0273	+0.0250	+0.0051	**+0.0148**
Convex Gating vs Per-Dim Gating							
Convex Gating	0.2866	0.6012	0.7439	0.8036	0.8418	0.8556	**0.8669**
Per-Dim Gating	0.3143	0.6316	0.7581	0.8196	0.8358	0.8586	**0.8592**
Δ (A−B)	-0.0277	-0.0304	-0.0141	-0.0160	+0.0060	-0.0030	**+0.0076**
High-Dimensional Embedding vs Low-Dimensional Embedding							
High-Dimensional Embedding	0.3408	0.6464	0.7596	0.8088	0.8453	0.8609	**0.8770**
Low-Dimensional Embedding	0.2995	0.6329	0.7459	0.8069	0.8425	0.8599	**0.8724**
Δ (A−B)	+0.0413	+0.0135	+0.0137	+0.0019	+0.0028	+0.0010	**+0.0045**

Figure 4: Comparison of architectural design choices. Validation $AUPR_{micro}$ curves and tabulated performance deltas illustrating the impact of key design variants across training epochs: SwiGLU + register tokens, embedding dimensionality, pooling strategy, gating type, and transformer configuration. Source: Table/Graphs created by the student researcher, 2025.

Experimental setup. To validate the design of our fusion architecture, we conducted systematic ablations of key architectural components on the molecular function prediction task. These experiments isolate the contribution of specific design choices while holding training procedures, data, and other hyperparameters constant. We evaluate variants spanning multiple dimensions of the architecture: (1) attention mechanisms—bidirectional cross-attention between streams versus independent self-attention within each stream; (2) pooling process—cross attention streamed pooling versus naive concatenated projection pooling for the initial two groupings of PLMs; (3) architectural enhancements—incorporating learnable register tokens with SwiGLU activations versus standard configurations without these components; and (4) gating strategies—per-dimension element-wise gating versus convex (softmax-normalized) gating for combining modality representations; (5) embedding dimensionality—utilizing higher-dimensional PLM features (Ankh3-XL at 2560d) versus lower-dimensional features (Ankh3-L at 1536d). All models are trained for seven epochs on identical data splits and evaluated using validation $AUPR_{micro}$ to assess ablation performance.

Results. The results demonstrate that the bidirectional cross-attention transformer consistently outperforms the self-attention transformer by enabling direct information exchange between modality streams, while cross-attention pooling provides lifts for similar reasons in earlier stages of the model. Convex gating provides more stable fusion than per-dimension gating by ensuring weighted combinations sum to unity (avoiding overfitting in the process). Higher-dimensional PLM embeddings yield modest but consistent gains, suggesting that preserving the full representational capacity of pretrained models is beneficial. The combination of learnable register tokens and SwiGLU activations improves performance through enhanced expressivity and better gradient flow. Collectively, these ablations demonstrate that our design choices are complementary and compound to form a well-calibrated architecture for multi-PLM protein representation learning. Additional experiments—covering broader training hyperparameters such as loss formulation (BCE vs. ASL, with ASL performing better), optimal latent dimensionality (896), weight-averaging strategy (EMA outperforming standard averaging), learning-rate schedules, optimizers, and related tuning parameters—were also explored but are omitted here for brevity.

6 Web Tool and Accessibility

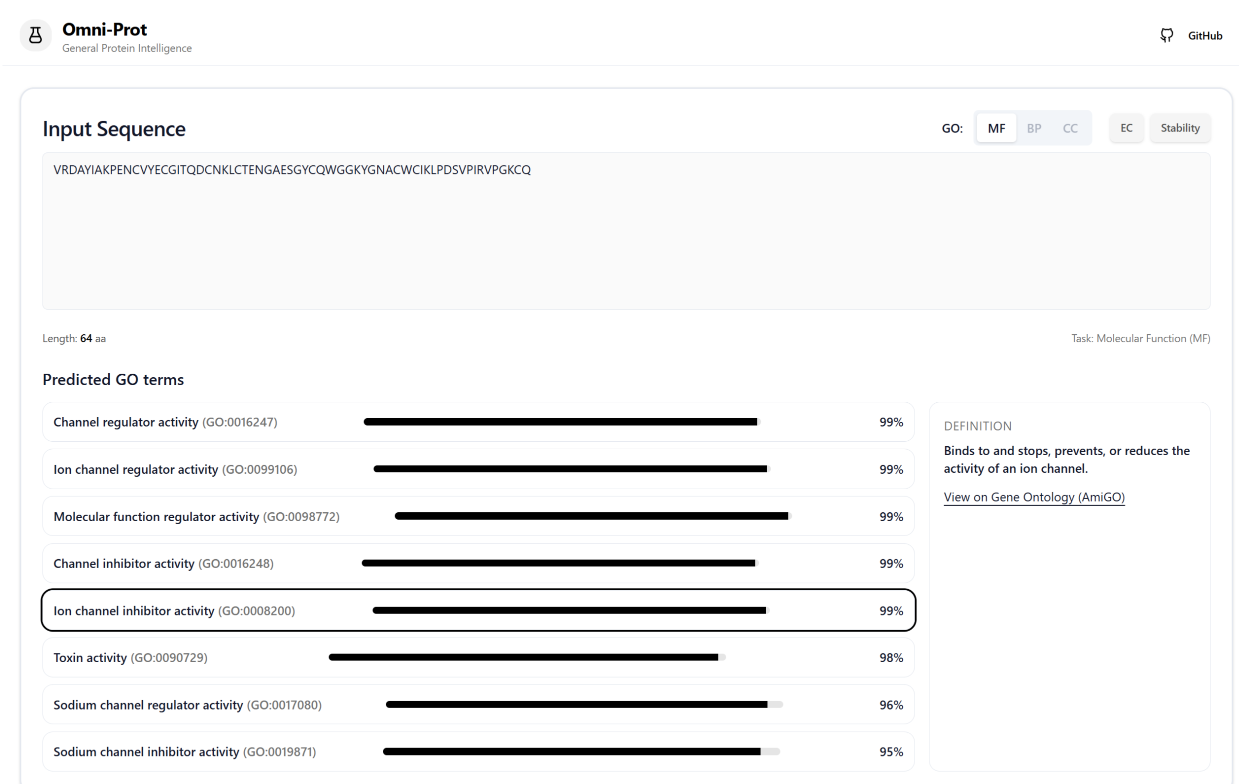


Figure 5: Omni-Prot web UI showing correct molecular function predictions for a held-out protein (PDB: 2KBK, chain A). The tool lists GO terms sorted by model probability and hyperlinks each to its official entry. Notable, rare predictions include *ion channel inhibitor activity* (GO:0008200, ~0.99) and *sodium channel inhibitor activity* (GO:0019871, ~0.95), with *toxin activity* (GO:0090729, ~0.98) providing a coherent mechanistic context. Definitions are retrieved via in-app AmiGO links. Backend: a serverless pipeline tokenizes the sequence, runs inference on relevant Omni-Prot weights, and returns predictions exceeding threshold τ. Source: Graphic created by the student researcher, 2025.

We release a user-friendly web application to make the model immediately accessible for practical use, designed primarily for GO term prediction where our approach shows the strongest performance improvements and is most useful. While several deep learning models for protein function prediction exist, many lack accessible web interfaces for practical deployment, limiting their use by the broader research community. Moreover, current web-based GO tools rely on BLAST and homology transfer, which cannot learn functional patterns beyond sequence similarity close database matches. Our model overcomes these limitations while providing massive improvements compared to existing homology-based methods (as shown in Table 1), making it particularly useful for annotating metagenomic discoveries, interpreting disease variants, and accelerating pathway design (within the supported GO dictionary). The tool also optionally supports EC classification and stability prediction for enzyme engineering applications. Besides improved performance, a key advantage our model enables is scalable mass annotation of proteins for large-scale analyses due to the model's sole reliance on amino acid, a critical capability for high-throughput proteomics and functional genomics studies. Furthermore, the modular architecture enables researchers to adapt the framework to new tasks through fine-tuning on custom datasets and to seamlessly integrate more powerful PLMs as they emerge, ensuring long-term relevance without architectural redesign.

7 Limitations

First, while preliminary experiments across multiple random seeds and datasets showed low variance in results, with the MF PDBch task serving as a strong proxy in ablations for most protein function tasks (Section 5), computational constraints prevented extensive repeated trials for all experimental configurations across all datasets. Second, even though it remains a prevalent benchmark in literature, the EC number classification dataset exhibits notable inconsistencies between training and test distributions including labels that were not present in the training set but existed in the test set. In addition, substantial label noise with many labels in the EC dictionary having a single or small few labels may hinder its effectiveness as a standardized benchmark compared to more carefully curated resources like PDBch or TAPE stability. This data quality issue complicates fair model comparison and may inflate or deflate apparent performance gains in the EC table. Third, predictive performance on proteins from deeply understudied taxonomic lineages, extreme environments, or synthetic designs that diverge substantially from natural evolutionary trajectories may vary. Future work should systematically evaluate robustness across these underrepresented protein spaces to establish confidence bounds for deployment in frontier applications.

8 Conclusion

In sum, Omni-Prot unifies complementary PLMs via reciprocal attention and gated fusion, achieving state-of-the-art on GO, EC, and stability. Ablations show that integrating diverse PLM representations through a targeted architecture drives performance, while MSA and 3D coordinates add little once strong sequence embeddings are utilized. This sequence-only framework streamlines deployment and broadens applicability, with our open implementation and simple web interface making these advances readily accessible.

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